Image Operations

Negative ($s = L-1-r$): Inverts all intensities. Dark becomes bright, bright becomes dark.

Log Transform ($s = c\log(1+r)$): Compresses high intensities, expands low ones. Reveals detail in dark regions.

Power-Law / Gamma ($s = cr^{\gamma}$): $\gamma < 1$: brightens dark regions. $\gamma > 1$: darkens image, suppresses low intensities.

Contrast Stretching: Maps a narrow intensity range to the full $[0, L-1]$ range. Increases overall contrast.

Intensity-Level Slicing: Highlights a specific intensity band while suppressing or preserving all others.

Bit-Plane Slicing: Decomposes into 8 binary planes. Higher planes carry structure; lower planes carry fine detail/noise.

Thresholding ($s = 0$ or $L-1$): Produces binary image. Separates foreground from background.

Histogram Equalization: Flattens histogram to use full intensity range. Increases global contrast.

Histogram Matching: Reshapes histogram to match a specified target distribution.

Local Histogram Processing: Applies equalization in sliding neighborhoods. Reveals local detail that global methods miss.

Addition / Averaging: $\bar{g}(x, y) = \frac{1}{K}\sum g_i(x, y)$. Reduces noise variance by $1/K$. Images must be registered.

Subtraction: Highlights differences between two images. Removes static background.

Multiplication / Division: Mask with ROI to isolate regions. Division corrects uneven illumination/shading.

Smoothing (Box / Gaussian): Averages neighborhood pixels. Reduces noise but blurs edges.

Median Filter (nonlinear): Replaces pixel with neighborhood median. Removes salt-and-pepper noise while preserving edges.

Sharpening (Laplacian, Unsharp Mask): Enhances edges and fine detail. Makes transitions crisper.

Scaling / Resizing: Requires interpolation—NN is blocky, bilinear is smooth, bicubic preserves detail best.

Rotation / Translation: Repositions image. Requires resampling via interpolation at non-integer coordinates.

NOT: swaps foreground/background. **AND**: intersection of two masks. **OR**: union of two masks. **XOR**: pixels in one mask but not both.